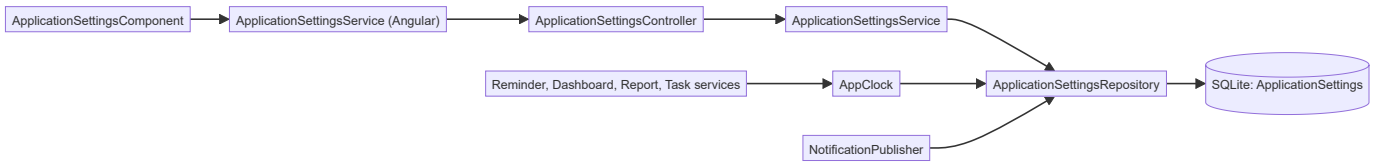
Application Settings – Technical Documentation

> **Project:** TaskFlow – Intelligent Task Allocation & Tracking System **Parent Module:** Settings · **Child Module:** 08.1 Application Settings **Version:** 1.0 · **Last Updated:** 31-Oct-2026 **Backend:** ASP.NET Core Web API (.NET 10) · **Frontend:** Angular · **Database:** SQLite (EF Core) **Author:** Jojo George (jojogeorge3344@gmail.com) © 2026 Jojo George. All rights reserved.

1. Technical Overview

Application Settings is `ApplicationSettingsController` over `ApplicationSettingsService` and the shared `ApplicationSettingsRepository`. It reads and writes four keys of the key/value table `ApplicationSettings` (`Application.Name`, `Application.Timezone`, `Application.DefaultPriority`, `Notification.InAppEnabled`). The GET endpoint is open to every signed-in user (brand name, default priority); the PUT endpoint needs `settings.application.manage`. `AppClock` (`IAppClock`) resolves the single application timezone used by all deadline-day calculations.

2. Architecture Flow



3. Backend File Structure

Layer	Path	Files
Controller	`TaskFlow.Api/Controllers/Settings/`	`ApplicationSettingsController.cs`
Service	`TaskFlow.Api/Services/Settings/ApplicationSettings/General/`	`IApplicationSettingsService.cs`, `ApplicationSettingsService.cs`
Service (Clock)	`TaskFlow.Api/Services/Settings/ApplicationSettings/Clock/`	`IAppClock.cs`, `AppClock.cs`
Repository (shared with 08.2)	`TaskFlow.Api/Repositories/Settings/ApplicationSettings/`	`IApplicationSettingsRepository.cs`, `ApplicationSettingsRepository.cs`
DTOs	`TaskFlow.Api/Dtos/Settings/ApplicationSettings/`	`ApplicationSettingsDtos.cs` (`ApplicationSettingsDto`, `UpdateApplicationSettingsRequest`)
Validators	`TaskFlow.Api/Dtos/Settings/ApplicationSettings/Validators/`	`UpdateApplicationSettingsRequestValidator.cs`
Entities	`TaskFlow.Api/Models/Entities/`	`ApplicationSetting.cs`
Enums	`TaskFlow.Api/Models/Enums/`	`TaskPriority.cs`
Constants	`TaskFlow.Api/Common/Constants/`	`SettingKeys.cs`

Layer	Path	Files
Seeders	`TaskFlow.Api/Persistence/Seeders/`	`SettingsInitializer.cs`
Config	`TaskFlow.Api/Config/`	`ApplicationOptions.cs`

4. Frontend File Structure

Type	Path	Purpose
Page component	`src/app/modules/settings/application-settings/pages/application-settings.component.{ts,html}`	`ApplicationSettingsComponent` (Screen 20)
Models	`src/app/modules/settings/application-settings/pages/application-settings.model.ts`, `src/app/modules/settings/models/settings.model.ts`	`SettingsField`, `ApplicationSettings` interfaces
API service	`src/app/modules/settings/application-settings/services/application-settings.service.ts`	`ApplicationSettingsService.get()/save()` and the cached `appName` signal used by the shell
Route / Guard	`src/app/modules/settings/settings.routes.ts` (`settings/application`), `src/app/shared/auth/auth.guard.ts`	`permissionGuard` with `SettingsApplication`
Menu	`src/app/layouts/nav-items.ts`	"Application Settings" item when `SettingsApplication` is held

5. API & Logic

5.1 API Endpoints

#	Method	Endpoint	Description	Permission (Policy)	Request DTO	Response DTO
1	GET	`/api/v1/settings/application`	Current application settings (non-secret)	`[Authorize]` only (any signed-in user)	–	`ApplicationSettingsDto`
2	PUT	`/api/v1/settings/application`	Save all application settings	`settings.application.manage`	`UpdateApplicationSettingsRequest`	`ApplicationSettingsDto`

5.2 Request / Response Samples

```
// PUT /api/v1/settings/application
{ "name": "TaskFlow", "timezone": "Asia/Kolkata", "defaultPriority": "Medium",
"inAppNotificationsEnabled": true }
```

```
// 200 OK
{ "name": "TaskFlow", "timezone": "Asia/Kolkata", "defaultPriority": "Medium",
"inAppNotificationsEnabled": true }
```

> Enums are serialised as strings (`JsonStringEnumConverter` in `Config/CrossCuttingRegistration.cs`).

5.3 Service Layer Logic

Method: `ApplicationSettingsService.GetAsync(CancellationToken)`

1. `GetValuesAsync` for keys `Application.Name`, `Application.Timezone`, `Application.DefaultPriority`, `Notification.InAppEnabled`.
2. Each value falls back to a default when missing or unparseable: `"TaskFlow"`, `ApplicationOptions.Timezone` (or `"UTC"`), `TaskPriority.Medium`, `true`.

Method: `ApplicationSettingsService.UpdateAsync(int userId, UpdateApplicationSettingsRequest request, CancellationToken)`

1. The controller resolves `userId` from `ClaimTypes.NameIdentifier`, or returns `401`.
2. If any field is null, throws `ArgumentException("Every application setting must be provided.")`. The validator normally prevents this.
3. `GetForUpdateAsync` loads the tracked rows. A missing key is created with its category (`Application` or `Notification`).
4. Sets `Value`, `UpdatedByUserId = userId` and `UpdatedAt = now` for all four keys. The name is trimmed, and the bool is written with `SettingValues.Format` (`"true"` / `"false"`).
5. `IUnitOfWork.SaveChangesAsync`, then returns `GetAsync()`.

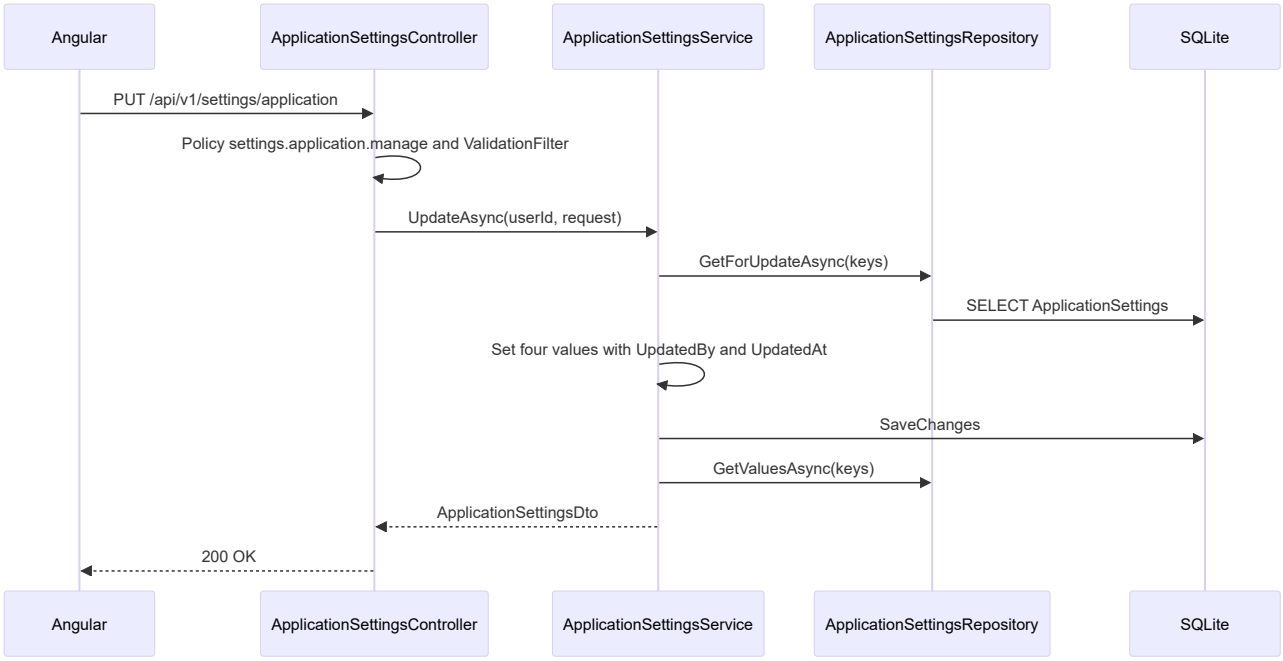
Method: `AppClock.GetTimeZoneAsync()` / `TodayLocalAsync()` (`IAppClock`)

- Reads `Application.Timezone` (falls back to `ApplicationOptions.Timezone`), then calls `TimeZoneInfo.FindSystemTimeZoneById`. An unknown id is logged as an error and `TimeZoneInfo.Utc` is returned. `TodayLocalAsync` converts `UtcNow` to that zone. It is used by the reminder, dashboard, workload, report, task and assistant services (BR-17).

5.4 Repository Layer

Method	Description	Query Notes
`GetValuesAsync(keys)`	Values of the given keys	`AsNoTracking`, `ToDictionaryAsync(Key, Value)`
`GetValueAsync(key)`	One value or null	`AsNoTracking`, projection
`GetForUpdateAsync(keys)`	Tracked rows for update	Tracked, `ToDictionaryAsync(Key)`
`Add(entity)` (base)	Stage a new row	From `BaseRepository`

5.5 Sequence Diagram



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6. Database Design

Table	Column	Type	Constraints	Description
`ApplicationSettings`	`Id`	int	PK, Identity	Row id
`ApplicationSettings`	`Key`	string(128)	Not Null, Unique	This child uses `Application.Name`, `Application.Timezone`, `Application.DefaultPriority`, `Notification.InAppEnabled`
`ApplicationSettings`	`Value`	string(1024)	Not Null	Invariant-culture string value
`ApplicationSettings`	`Category`	string(64)	Not Null	`Application` or `Notification` for this child
`ApplicationSettings`	`UpdatedByUserId`	int?	– (no FK)	Last editor
`ApplicationSettings`	`UpdatedAt`	DateTime	Not Null	UTC

Relationships: none (key/value table, no FK). The table is shared with 08.2 Reminder Settings.

Seed data: `SettingsInitializer` inserts missing setting rows at every start and never overwrites existing ones. `Application.Timezone` is seeded from `Application:Timezone`.

7. Security & Authorization

- **Authorization:** `PUT` needs `[RequirePermission(settings.application.manage)]`. `GET` is `[Authorize]` only, because the shell brand name and the Create Task default priority are needed by every user. The response contains nothing secret.
- **Data scope:** None. The settings are organization-wide.

- **Frontend:** `permissionGuard` on `settings/application` and the menu item in `buildNavigation()`.

Permission Key	Endpoint(s)	UI Element Controlled
`settings.application.manage`	`PUT /api/v1/settings/application`	Menu "Application Settings", route `/settings/application`
(signed in)	`GET /api/v1/settings/application`	Shell brand name, Create Task default priority

8. Validations & Error Handling

Rule	Layer	HTTP Code	Error Message
Application name blank / > 100	`UpdateApplicationSettingsRequestValidator`	400	Application name cannot be blank. (length: FluentValidation default)
Timezone blank	Validator	400	Timezone cannot be blank.
Timezone not found by `TimeZoneInfo.FindSystemTimeZoneById`	Validator	400	Unknown timezone.
DefaultPriority / InAppNotificationsEnabled missing or invalid	Validator	400	FluentValidation default

Shared rules (403, 401, 409 concurrency, 500) are in the parent document. The screen maps the `errors` keys onto the form controls.

9. Integrations / Side Effects

- **Notifications:** `NotificationPublisher` checks `Notification.InAppEnabled` before it creates any bell notification. A missing or unreadable value counts as enabled.
- **Timezone:** `IAppClock` gives the local calendar day to the reminder, dashboard, workload, report, task and assistant services.
- **AI Assistant:** `AssistantService` and `AssistantController` use `IApplicationSettingsService` for the brand name. Assistant knowledge page: `Services/Assistant/Knowledge/application-settings.md`.
- **Activity Log:** none (only `UpdatedByUserId` / `UpdatedAt`).

10. Configuration

Key (`appsettings.json`)	Purpose	Default
`Application:Timezone`	Seeds `Application.Timezone`. Fallback while the row is missing	`Asia/Kolkata` (code default `UTC`)

Runtime-editable keys in the `ApplicationSettings` table (`SettingKeys`):

- `Application.Name` (`TaskFlow`), `Application.Timezone`, `Application.DefaultPriority` (`Medium`)
- `Notification.InAppEnabled` (`true`)

11. Testing

Test Type	File / Location	Covers
Unit	`tests/TaskFlow.UnitTests/Settings/ApplicationSettings/General/ApplicationSettingsServiceTests.cs`	Defaults on a fresh database; missing or unreadable rows fall back to the defaults and the configured timezone; saved values round-trip with who and when; an incomplete request throws and saves nothing
Unit	`tests/TaskFlow.UnitTests/Settings/ApplicationSettings/General/ApplicationSettingsValidatorTests.cs`	Defaults are valid; a blank name, an unknown timezone, missing fields and a priority outside the enum are invalid
Unit	`tests/TaskFlow.UnitTests/Settings/ApplicationSettings/Clock/AppClockTests.cs`	Uses the stored IANA timezone, or the configured one while none is stored; an unknown timezone falls back to UTC; "today" is the local date in the application timezone
Integration	`tests/TaskFlow.IntegrationTests/Settings/ApplicationSettings/ApplicationSettingsEndpointsTests.cs`	`GET` works for Employee, Manager and Administrator; the Administrator's `PUT` saves and returns the values; a Manager's `PUT` gets 403; an unknown timezone gets 400 with a `Timezone` field error and saves nothing; a blank name gets 400; no token gets 401
Frontend unit	`src/app/modules/settings/application-settings/pages/application-settings.component.spec.ts`	Loads and shows the stored values; saves and confirms; blocks a blank name; shows server field errors from a 400; hides "Saved" after an edit; timezone suggestions via a datalist that still allows custom ids; links to Reminder Settings; retryable load error
Frontend unit	`src/app/modules/settings/application-settings/services/application-settings.service.spec.ts`	`GET` and `PUT` on `/api/v1/settings/application`; the cached brand name defaults to `TaskFlow`, updates after a read or save, and falls back to `TaskFlow` when the name is blank

> Shared tests (`SettingsInitializerTests.cs`) are in the parent document.

12. Known Limitations / Gaps found

- **BRD Screen 20 "Default task reminder intervals"** are not on Application Settings. The screen links to Reminder Settings instead (by design, see the component comment).
- **Stale comments:** `ApplicationSettingsService` (Angular) and `ApplicationSettingsController` say "only an Administrator", but the code checks `settings.application.manage`.
- Module-wide limitations (no audit log, no reset to defaults) are in the parent document.

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